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The Effects of Reward Contingency and Performance Feedback on Intrinsic Motivation

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In this study the effects of reward contingency and positive performance feedback on subsequent intrinsic motivation for an enjoyable task were examined. It was hypothesized that rewards contingent only upon participation in the task would produce decrements in motivation (the overjustification effect) but that rewards contingent on performance quality would produce even greater decrements in intrinsic motivation relative to control conditions of no reward. It was also hypothesized that positive performance feedback would enhance intrinsic motivation and that this effect would be independent of reward effects. High school students were offered performance or task-contingent rewards, or no reward, for doing hidden-figures puzzles. Subjects offered performance-contingent rewards all received positive feedback concerning performance and half the subjects in task-contingent and no-reward conditions received the same positive feedback. Performance-contingent rewards were found to undermine intrinsic motivation more than task-contingent ones, which produced decrements relative to control conditions of no reward, supporting Deci's control model. Positive feedback enhanced intrinsic motivation and this effect was independent of reward effects. A recall measure indicated that subjects receiving performance-contingent rewards remembered fewer performance-irrelevant details about the task, suggesting that rewards may affect the process of task involvement as well as its motivational outcomes.

In a review of the literature concerned with the effects of extrinsic rewards on intrinsic motivation, Condry (1977) concludes that in certain contexts, subsequent interest in a task may be reduced by the imposition of task-extrinsic rewards. One aspect of context that apparently mediates this undermining effect is

reward contingency. Rewards have been promised to subjects contingent only upon participation or for completion of the task (task contingent),¹ or contingent upon some level of performance of the task (performance contingent).

The undermining effects of task-contingent rewards have been well documented (Ander-son, Manoogian, & Reznick, 1976; Lepper, Greene, & Nisbett, 1973). Rewards promised to subjects for engaging in an activity and implying no performance demands have consistently produced what Lepper and Greene (1976) termed the "overjustification effect,"

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¹ These rewards have frequently been referred to as *noncontingent* rewards. Although not explicitly contingent on some level of performance, they are usually at least implicitly contingent on compliance with the experimenter or participation in the activity and are perhaps more appropriately considered task contingent.

or a decrease in subsequent intrinsic motivation. Their most recent theoretical statement specifies the context in which this effect should occur:

The overjustification studies provide a demonstration of the negative effects on intrinsic motivation of making salient to a person the instrumentality of his behavior when rewards are presented which do not result in the acquisition of new skills or convey salient information to him concerning his ability at a task. (Lepper & Greene, 1976, p. 33)

Because the promise of a performance-contingent reward implies that information concerning task ability will be provided, specific predictions for the effects of performance-contingent rewards cannot be directly derived from the overjustification hypothesis. Other investigators, however, have attempted to examine the effects of such rewards and to relate them to overjustification effects. Two predictions for performance-contingent rewards have been advanced. Karniol and Ross (1977) suggested that performance-contingent rewards may maintain or increase intrinsic motivation to the extent that they convey information about effective performance at a task, because attainment of such rewards should provide tangible evidence of personal effectiveness. Deci (1975), on the other hand, predicted that a performance-contingent reward should decrease intrinsic motivation even more than a task-contingent one because a reward is perceived to be more controlling when it is contingent on some level of performance.

Karniol and Ross and Deci agreed that rewards have controlling and informational aspects, and both suggested that the more salient of the two will initiate either changes in perceptions of instrumentality of behavior (when controlling aspects of the reward are salient) or changes in feelings of competence and self-determination (when informational aspects are salient). Their predictions for the effects of performance-contingent rewards differed because of differences in the ascribed salience of controlling versus informational aspects of these rewards.

Previous Research

Deci (1971, 1972a) consistently found that subjects receiving performance-contingent re-

wards demonstrate less subsequent intrinsic motivation than unrewarded controls. Pritchard, Campbell, and Campbell (1977) and Pittman, Cooper, and Smith (1977) found that performance-contingent rewards undermined intrinsic motivation relative to control conditions of no reward. Pinder (1976) found that performance-contingent rewards produced greater decrements in intrinsic motivation than did task-contingent ones. Although these results all seem to disconfirm the hypothesis of Karniol and Ross, the studies do not represent adequate tests of it, because performance-contingent rewards are predicted to maintain interest only when subjects perceive that they have done well at the task. It is not clear that this criterion was met in these studies.

Two studies have compared task-contingent and performance-contingent rewards with a control condition of no reward in the same design. Greene and Lepper (1974) found that performance and task-contingent rewards produced equal decrements in intrinsic motivation relative to control conditions. Karniol and Ross (1977) found that when children received positive feedback concerning their performance, task-contingent rewards decreased subsequent interest relative to performance-contingent rewards and no rewards. Performance-contingent rewards did maintain interest, but they did not increase it above the control level as had been predicted. Karniol and Ross suggested that because the performance standard was established in the reward manipulation at the outset, children knew that they had performed well as soon as they finished the task. The information inherent in the receipt of the reward was thus redundant with that provided in the instructions. The researchers predicted that when the receipt of a reward provides additional rather than redundant information concerning competence, it should increase intrinsic motivation above the level of control subjects.

Methodological Issues

One aspect of reward contingency that has not been directly examined is the possible confounding of material reward with positive

feedback about performance quality. The promise of a performance-contingent reward implies that a subject's performance will be evaluated and that feedback will be available. Deci (1972b) and Anderson et al. (1976) demonstrated that positive feedback, in the absence of any reward manipulation, increases subsequent intrinsic motivation. Although it is difficult to separate reward manipulation from feedback in the case of performance-contingent rewards, it is possible to examine the effects of reward and feedback independently in task-contingent and control conditions because feedback is not implied by a task-contingent reward manipulation.

Several measures of intrinsic motivation have been used in the overjustification literature, ranging from behavioral measures (task persistence in the absence of rewards) to volunteering measures (willingness to return for another session) to self-report measures of task interest and enjoyment. Conflicting results of previous studies may be due to different operational definitions of intrinsic motivation.

Condry (1977) draws a distinction between dependent measures that focus on the products of the rewarded session, for example, subsequent interest, and measures that focus on the process of task involvement, and he suggests that the promise of extrinsic rewards may affect the manner in which subjects engage in a task, as well as subsequent motivation. Products and processes may be differentially affected by rewards, however, and should be considered separately.

Objectives

The primary aim of the present study was to examine the effects of performance- and task-contingent rewards on intrinsic motivation, compared to conditions of no reward. To accomplish this, two kinds of performance-contingent rewards were created. A performance-contingent reward with performance norms supplied at the outset of the task would convey no additional information concerning competence beyond that inherent in the norms. A performance-contingent reward with no prior norms established would provide

nonredundant information concerning competence. These two performance-contingent rewards were compared with task-contingent rewards and control conditions of no reward. In addition, the effects of positive performance feedback, relative to no feedback, were examined in conditions of task-contingent and no reward in order that results might be comparable to both those of Karniol and Ross and those of other studies that have demonstrated the undermining effects of task-contingent rewards in the absence of positive performance feedback.

To facilitate comparison with other studies, several behavioral, volunteering, and self-report product measures of intrinsic motivation were utilized. Interest was measured at several points in time in order to examine the endurance of reward effects. The task in this study was constructed in such a way that attention to aspects of it not relevant to performance could be measured with an incidental recall measure, intended to assess the process of intrinsic motivation.

It was predicted that positive feedback would enhance intrinsic motivation, that task-contingent rewards would undermine it, and that these two effects would be independent. The effects of performance-contingent rewards were of particular interest because of conflicting predictions: an enhancement of motivation mediated by perceived self-efficacy (Karniol & Ross, 1977) or a detrimental effect because of the perceived controlling nature of the reward (Deci, 1975). Controlling the informational value of the reward permits closer analysis of performance-contingent rewards. Karniol and Ross predict that the more informational performance-contingent reward with no prior norms established should produce the most enhancement of motivation. Although Deci does not make specific predictions regarding informational value, a speculative control prediction was generated. Because the two rewards are both contingent on performance quality, equal decrements might be predicted, but this negative effect could be differentially mitigated by the two types of positive feedback. When norms are supplied at the outset, the feedback is received earlier in the motivational process and might have a

greater positive effect than would feedback received after completion of the task. According to this analysis, the control hypothesis would produce a larger decrement in interest in the performance-contingent condition where norms are not supplied. It was hypothesized that the control prediction would be supported in this study because of the greater similarity of the subject population (high school students) to Deci's college students than to the children used as subjects by Karniol and Ross.

Method

Overview

In this study six groups of subjects received various combinations of reward and positive feedback with respect to a challenging but enjoyable task. Interest in this task was assessed both before and after the experimental manipulations. The six experimental conditions are summarized in Table 1. The no reward–no feedback (NRNF) and task-contingent reward–no feedback (TCRNF) cells replicate the conditions employed in overjustification studies, whereas the no reward–positive feedback (NRPF), task-contingent reward–positive feedback (TCRPF) and performance-contingent reward (norms supplied)–positive feedback (PCRNPF) cells are conceptual replications of the central features of the Karniol and Ross (1977) design. Subjects in the PCRNPF condition were offered a reward contingent on performing

above average and were informed of the performance standard before beginning the task, so that receipt of the reward only confirmed that they had performed above average. In the performance-contingent reward (no norms)–positive feedback (PCRNPF) condition, however, the reward itself was the only source of information available to the subject concerning level of performance. Thus the reward received in the PCRNPF condition was more informational than that received in the PCRNPF condition, although subjects in both conditions received the same feedback concerning performance. The standard of performance was set at a relatively low level so that all subjects would be eligible for positive feedback.

Subjects

The participants in this study were 64 male and 29 female students in a suburban Massachusetts high school, with a mean age of 16.4 years. The subjects were tested during their English class.

Procedure

The study was conducted in three phases. The first and third sessions were group administered in the classroom, whereas subjects were tested individually in a separate room for the second session. This experimental session began 1 month after the first session and continued for 2 weeks, and the third session took place 1 month after the end of the experimental sessions.

The experimental activity in this study was a series of hidden-figure puzzles—cartoon-style drawings by Al Hirschfeld, in which the name *Nina* is hidden sev-

Table 1
Schematic Design of Experiment and Contrast Weights for Main Hypothesis

Condition	Reward	Feedback	Contrast weight
No reward–no feedback (NRNF)	None mentioned	None	1
No reward–positive feedback (NRPF)	None mentioned	Positive ^a	3
Task-contingent reward–no feedback (TCRNF)	Offered for doing puzzles	None	–1
Task-contingent reward–positive feedback (TCRPF)	Offered for doing puzzles	Positive ^a	1
Performance-contingent reward (norms supplied)–positive feedback (PCRNPF)	Offered for meeting performance criterion, norms supplied before task	Positive ^a	–1
Performance-contingent reward (no norms)–positive feedback (PCRNPF)	Offered for meeting performance criterion, no norms supplied	Positive ^a	–3

^a Positive feedback was "We've found that the average student usually finds 4 Ninas, so you did better than the average high school student on these puzzles."



Figure 1. A Nina puzzle (Hirschfeld's drawing for the play *Anna Christie*, © courtesy of the Margo Feiden Galleries, New York, New York).

eral times. These drawings appear in the Arts and Leisure section of the Sunday *New York Times* and are usually caricatures of actors and scenes from current films and plays. Each drawing has a title and a caption describing the scene and naming the actors depicted. The drawings used in this study contained three to seven hidden Ninas. An example of a Nina puzzle (in which there are seven hidden Ninas) is shown in Figure 1.

Pretest. The students were given booklets containing three kinds of puzzles. The word puzzles con-

sisted of several anagrams; the face and body recognition puzzle was the booklet form of the Profile of Nonverbal Sensitivity (PONS; Rosenthal, Hall, DiMatteo, Rogers, & Archer, 1979), which consists of 40 pictures of a face or body and two brief situation labels from which the subject selects the more appropriate description of the picture; there were also three Nina puzzles. The order in which students did the three puzzle sets was counterbalanced. After doing each puzzle set, students completed a puzzle questionnaire that consisted of 13 7-point bipolar

rating scales (e.g., good-bad, easy-hard, boring-exciting) and eight questions concerning the subject's involvement with the puzzle (e.g., "How much did you enjoy the puzzle?"), also answered on 7-point scales. After completing all three types of puzzles, the students rank ordered them according to enjoyment.

Experimental session. The experimenter informed each subject that he or she would be doing more Nina puzzles and that the instructions were tape recorded. After turning on the tape recorder she left the room in order to remain blind to the subject's condition, which was randomly determined. All subjects were instructed to find as many Ninas as possible within the 2 minutes of inspection time allowed per drawing. Subjects in the two task-contingent reward conditions (TCRNF and TCRPF) were promised a reward for doing the puzzles, whereas subjects in the two performance-contingent reward conditions (PCRNSPF and PCRNPF) were promised a reward for exceeding the performance of the average high school student. Those in the performance-contingent reward (norms supplied) – positive feedback group were then informed of the exact performance standard (detecting four or more Ninas in the three puzzles combined). All subjects who had been promised a reward (TCRNF, TCRPF, PCRNSPF, and PCRNPF) were told that it consisted of two felt-tip pens and a notebook, in their choice of colors, and were reminded of the conditions of the reward. Finally, the subjects completed an identification form and answered a series of questions concerning their understanding of the directions, the reward contingencies, their current mood state, expected puzzle difficulty and performance level, and attitude toward the reward.

The experimenter returned to the room and administered the three Nina puzzles. Subjects in the NRPF, TCRPF, PCRNSPF, and PCRNPF groups then received positive performance feedback ("We've found that the average high school student usually finds four Ninas, so you did better than the average student on these puzzles"), and those who were promised a reward (TCRNF, TCRPF, PCRNSPF, and PCRNPF) indicated their color choices on forms that again reminded them of the conditions under which the reward had been promised. The experimenter was blind to this latter information. While the experimenter filed away the puzzles and order forms, the subject was permitted to read a current popular magazine, work on another Nina puzzle, or simply sit quietly. The experimenter unobtrusively observed the subject's behavior and recorded the amount of time spent on the puzzle, if any.

After 2 minutes the subject completed several questionnaires, including a Nina puzzle questionnaire identical to the one administered at the pretest. A volunteering questionnaire stated that although there were no definite plans for future sessions, the researcher wanted a general idea of student willingness to return. It was composed of four 7-point rating scales asking the subject how willing he or she would be to return for a further, unrewarded session of Nina puzzles if it were scheduled during their English class, during

a study hall, after school, or on a weekend. The extra puzzles request form informed subjects that copies of different Nina puzzles would be available after the conclusion of the research and permitted students to request up to four additional puzzles. The recall questionnaire was composed of a list of actors, in which were embedded the names of those depicted in the Nina puzzles from the experimental session, and a similar list of play and film titles; students were asked to circle any names that they remembered seeing in the puzzles.

If the subject had not previously been offered a reward, he or she was informed at this time that there was a gift for having participated in psychological research. These subjects then indicated their color choice for the pens and notebook on an order form that reminded the subject of the rationale for the gift.

Finally, all subjects were assured that the experiment would be explained to them in detail when the rewards/gifts were delivered at its conclusion. They were also cautioned not to discuss the details of the experiment with other students.

Posttest. One month after all students had participated in the experimental session, their teacher distributed a letter from the experimenter saying that she would be visiting their classroom soon with deliveries and to answer questions about the research project. At this time they completed yet a third Nina puzzle questionnaire.

The experimenter visited the classroom a few days after the students received the letter, distributed the rewards/gifts (and extra puzzles if requested), and debriefed the students.

Dependent Measures

In addition to various manipulation checks concerned with the informational effects of positive feedback and postmanipulation motivation, several dependent variables were of interest in this study. Performance was indexed by the number of Ninas actually found on the three puzzles administered during the experimental session; all subjects found at least 4 Ninas, and no subject found all 15 contained in the drawings. Initial levels of intrinsic motivation were indexed by a pretest enjoyment scale derived from the pretest Nina puzzle questionnaire in the manner described below.

There were five outcome measures of intrinsic motivation taken after the experimental manipulation: (a) *experimental enjoyment*—a scale identical to the pretest enjoyment scale, but administered after the experimental task; (b) *time*—the amount of time that the subject spent looking at the extra Nina puzzle during the free-choice period; (c) *volunteering*—score on the volunteering questionnaire; (d) *extra puzzles*—the number of puzzles requested by the subject on the extra puzzles request form; and (e) *post-test enjoyment*—a scale identical to the pretest and experimental enjoyment scales, but administered dur-

ing the posttest.² A further measure was a process variable intended to index the level of task orientation during the experimental session: (f) *incidental recall*—the number of names and titles correctly recognized on the recall questionnaire, corrected for overcircling by subtracting one half the number of incorrect answers.

Results

Pretest Analyses

A principal-components factor analysis with varimax rotations was performed on the 21 items of the Nina puzzle questionnaire, and the first factor extracted accounted for 50.6% of the variance. Seven items had factor loadings higher than .70 and were considered to contribute to the definition of the factor: Enjoyable, Good, Interesting, Exciting, Enjoyment of Puzzle, Interest in Puzzle, and Worthwhile. This factor was interpreted independently by two judges to measure task enjoyment, and scores on each of the 7 items were combined using unit weighting to form an enjoyment scale. Pretest enjoyment was used as a covariate in subsequent analyses of covariance.

Since the overjustification effect is predicted to occur only when intrinsic motivation for an activity is initially high (Lepper & Greene, 1976) it is necessary to establish that the Nina puzzles were in fact intrinsically interesting. The mean score for pretest enjoyment was 33.98 ($SD = 9.20$); this was significantly higher than 25, the midpoint of the enjoyment scale, $t(92) = 9.41$, $p < .001$, $d = 1.95$, suggesting that subjects rated the Nina puzzles more enjoyable than not. Examination of the rank ordering of puzzles indicated that 50% of the subjects rated the Nina puzzles most enjoyable of the three. This observed frequency was significantly higher than the chance expectation of 33.33%, $\chi^2(1) = 10.88$, $p < .002$, $\phi = .34$, offering additional evidence for the high level of initial enjoyment of the Nina puzzles.

Manipulation Checks

The effectiveness of the positive feedback manipulation was examined in two Treatment $\times$ Sex (6×2) analyses of covariance³ covarying pretest enjoyment on the perceived

performance item ("How well do you think you did on these puzzles?") from the Nina puzzle questionnaire for the experimental session and posttest. A planned contrast tested whether subjects in the four conditions receiving positive feedback (NRPF, TCRPF, PCRNPF, and PCRNPF) were higher in perceived performance than subjects in the two no-feedback conditions (NRNF, TCRNF). In this contrast the first four groups were assigned weights of +1 and the latter two groups were given weights of -2. Subjects who received positive feedback showed higher levels of perceived performance both immediately following the experimental session, $F(1, 86) = 7.08$, $p < .01$, $\eta = .28$, and at the posttest 1 month later, $F(1, 80) = 7.03$, $p < .01$, $\eta = .28$.

A Treatment $\times$ Sex (6×2) analysis of variance was performed on the seven postmanipulation scales. No significant effects were found for alert, comfortable, relaxed, motivated, excited, anticipated puzzle difficulty, or expected performance level, suggesting that there were no treatment differences in motivation as measured by these scales after the students heard the instruction tape but before they actually did the puzzles.

In order to examine possible differences between the two groups that received performance-contingent rewards, post hoc protected t tests were used to compare the PCRNPF and PCRNPF groups on the seven postmanipulation scales and on experimental and posttest perceived performance. PCRNPF subjects were

² Six subjects graduated from high school before the administration of the posttest, so posttest enjoyment scores were not obtained for them.

³ Sex of subject was included as a factor in all analyses of variance, although it was not relevant to the main hypotheses of this study. A significant sex effect for posttest perceived performance was found, $F(1, 74) = 4.33$, $p < .05$, $\eta = .24$, indicating that males thought they had performed better than females. A significant sex effect for expected performance indicated that males thought they would do better than females on the puzzles, $F(1, 80) = 6.78$, $p < .05$, $\eta = .27$. A significant main effect for sex on recall, $F(1, 80) = 6.62$, $p < .05$, $\eta = .28$, showed that males were higher in amount of incidental recall than females, suggesting that the females may have been more task oriented. The sex differences found in this study were unanticipated, and interpretation of them would require replication.

significantly higher than PCRNPF subjects on expected performance level ("How well do you think you will do on these puzzles?"), alert, and posttest perceived performance (PCRNPF $M_s = 5.13, 4.93, \text{ and } 5.84$ vs. PCRNPF $M_s = 4.16, 3.69, \text{ and } 4.72$ on 7-point scales), $t_s(87) = 2.36, 2.46, \text{ and } 2.32$, respectively, all $p_s < .05$. The two groups did not differ significantly on comfortable, relaxed, motivated, excited, anticipated puzzle difficulty, or experimental perceived performance. This pattern of significant differences suggests that the groups differed primarily in expectations for and perceptions of performance level.

Intercorrelation of the Dependent Measures

Table 2 presents the correlation matrix for the six measures of intrinsic motivation (pretest, experimental, and posttest enjoyment, time, volunteering, and extra puzzles), performance, recall, and experimental and posttest perceived performance. The average intercorrelation of the six measures was .51 ($p < .001$), and each individual correlation was significant at the $p < .001$ level, indicating that all of the operationalizations of intrinsic motivation were positively interrelated.

Performance was positively related to pretest and experimental enjoyment, $r(91) = .20$ and $.22$; $p < .05$, but was not significantly related to any other measures of intrinsic motivation. Performance (actual) did correlate positively with experimental perceived performance, $r(91) = .33$, $p < .001$. Recall did not correlate significantly with any measures of intrinsic motivation or performance.

Because the four intrinsic motivation variables measured during the experimental session were positively related, a principal-components factor analysis was performed on them in order to create one measure of motivation for the experimental session. Volunteering, extra puzzles, time, and experimental enjoyment had loadings of .57, .75, .60, and .73, respectively, on the first factor extracted, and standardized factor scores were computed for the resulting intrinsic motivation measure, which was used as the primary dependent variable in the main analyses.

Effects of Reward and Feedback on Intrinsic Motivation

A Treatment $\times$ Sex (6×2) analysis of covariance with pretest enjoyment as a covariate was performed on intrinsic motivation

Table 2
Intercorrelations of Dependent Measures

Measure	1	2	3	4	5	6	7	8	9
1. Pretest enjoyment									
2. Experimental enjoyment	.69***								
3. Posttest enjoyment	.55***	.69***							
4. Time	.33***	.44***	.42***						
5. Volunteering	.37***	.50***	.40***	.33***					
6. Extra puzzles	.46***	.54***	.41**	.53***	.43***				
7. Performance	.20*	.22*	.14	.10	.09	.03			
8. Recall	.05	.04	.01	.14	.06	.17	.11		
9. Experimental perceived performance	.04	.11	.00	.02	.02	-.07	.30**	-.13	
10. Posttest perceived performance	.12	.19	.28**	.29**	-.02	.03	.33**	.08	.51***

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 3
Reward Condition Means for the Dependent Variables

Measure	Reward condition/contrast weight					
	NRPF/ +3	NRNF/ +1	TCRPF/ +1	TCRNF/ -1	PCRNPF/ -1	PCRNPF/ -3
Intrinsic motivation	1.05	-.16	.39	-.64	.07	-.96
Posttest enjoyment	38.47	33.09	35.36	31.96	35.97	29.96
Recall	2.56	2.44	3.13	2.30	1.49	1.22
Performance	8.87	7.81	8.56	8.27	9.00	8.62

Note. For the intrinsic motivation measure, standardized factor scores were created from volunteering, extra puzzles, time, and experimental enjoyment variables. NRPF = no reward - positive feedback. NRNF = no reward - no feedback. TCRPF = task-contingent reward - positive feedback. TCRNF = task-contingent reward - no feedback. PCRNPF = performance-contingent reward (norms supplied) - positive feedback. PCRNPF = performance-contingent reward (no norms) - positive feedback.

and posttest enjoyment. The effect of the pretest enjoyment covariate was highly significant for both intrinsic motivation, $F(1, 86) = 56.98$, $p < .001$, $\eta = .63$, and posttest enjoyment, $F(1, 80) = 39.99$, $p < .001$, $\eta = .58$, indicating that, as expected, intrinsic interest following the experimental manipulation was linearly related to initial task interest. No effects involving sex were significant for either dependent measure.

A planned contrast was used to test the main effect of treatment. Weights for the contrast were operationalized according to the control hypothesis as follows: no reward - no feedback—+1; no reward - positive feedback—+3; task-contingent reward - no feedback—-1; task-contingent reward - positive feedback—+1; performance-contingent reward (norms supplied) - positive feedback—-1; and performance-contingent reward (no norms) - positive feedback—-3. These equal interval weights are based on the assumption that the predicted effects are all equal in magnitude. This contrast tests the prediction that task-contingent rewards will produce negative effects relative to control conditions of no reward, that positive feedback will produce positive effects relative to conditions of no feedback, and that these two effects do not interact. This contrast also tests the prediction that performance-contingent rewards will decrease intrinsic motivation more than task-contingent ones and that performance-contingent rewards with no norms supplied will decrease intrinsic motivation more than per-

formance-contingent rewards with norms supplied.

Means for the two dependent measures of intrinsic motivation are presented in Table 3. As predicted, the treatment planned contrast was highly significant for both measures: intrinsic motivation, $F(1, 86) = 12.72$, $p < .001$, $\eta = .36$, and posttest enjoyment, $F(1, 80) = 9.58$, $p < .005$, $\eta = .33$.⁴ Thus intrinsic motivation was enhanced by positive feedback and undermined by rewards, particularly performance-contingent ones. This treatment planned contrast tests the overall patterning of the six cell means with respect to the predictions specified by the weights. Because this overall test incorporates several specific predictions, and since particular cell comparisons are relevant with respect to comparison with other studies, several additional nonorthogonal but theoretically relevant planned comparisons were computed for the primary dependent measure, intrinsic motivation.

In the four cells in which task-contingent reward or no reward was crossed with positive feedback or no feedback (TCRNF, TCRPF, NRNF, and NRPF), task-contingent rewards were found to reduce intrinsic motivation relative to control conditions of no reward, replicating the overjustification effect: TCRNF and

⁴ This planned contrast was also significant for each of the four dependent measures combined to form the intrinsic motivation measure, in four separate analyses of variance on time, extra puzzles, volunteering, and experimental enjoyment.

TCRPF versus NRNF and NRPF, $t(87) = 1.78$, $p < .05$, one-tailed,⁵ $d = .38$; and positive feedback increased interest relative to no feedback: NRPF and TCRPF versus NRNF and TCRNF, $t(87) = 2.74$, $p < .05$, $d = .59$. The interaction between task-contingent reward and feedback was not significant. Comparison of the two effect sizes (ds) suggests that the positive feedback effect was stronger than the overjustification effect.

Planned comparisons were also computed for the four cells with positive feedback (NRPF, TCRPF, PCRNPF, and PCRNPF), which represent the conceptual replication and extension of the Karniol and Ross (1977) design. The TCRPF and PCRNPF means were not significantly different, but both were significantly lower than the control group: TCRPF and PCRNPF versus NRPF, $t(87) = 2.01$, $p < .05$, $d = .43$. Thus the Karniol and Ross results were not replicated: Performance-contingent rewards, as well as task-contingent ones, reduced interest relative to control conditions. Furthermore, the Karniol and Ross prediction for the more informational PCRNPF group was not supported; the PCRNPF group was significantly lower than the PCRNPF group in interest, $t(87) = 1.76$, $p < .05$, one-tailed, $d = .37$. Additionally, although the PCRNPF group was not significantly lower than the TCRPF group, PCRNPF rewards were significantly more undermining than TCRPF ones, $t(87) = 2.34$, $p < .05$, $d = .50$, and the two performance-contingent rewards considered together were significantly more undermining than the task-contingent one: PCRNPF and PCRNPF versus TCRPF, $t(87) = 1.67$, $p < .05$, one-tailed, $d = .36$, replicating the results of Deci (1972b).

Effects of Treatment on Performance and Recall

Treatment $\times$ Sex (6×2) analyses of covariance covarying pretest enjoyment were performed on incidental recall and performance. There were no significant effects for performance.

It was predicted that subjects offered performance-contingent rewards would be more task oriented and would recall fewer irrelevant

details than subjects offered no rewards or task-contingent ones. A planned contrast comparing the two performance-contingent conditions (PCRNPF and PCRNPF: -2) with the other four conditions (NRNF, NRPF, TCRNF, and TCRPF: $+1$) was significant, $F(1, 86) = 8.95$, $p < .005$, $\eta = .32$, indicating that recall was lower in the two performance-contingent conditions, as predicted. The effect of the pretest enjoyment covariate was not significant ($F < 1$). Amount of recall, then, did not vary with the subjects' initial enjoyment of the puzzles. No other effects, except those involving sex, were significant for this measure.

Discussion

The control hypothesis received very strong support in this study. Performance-contingent rewards, particularly informational ones, were found to undermine subsequent intrinsic motivation more than task-contingent ones, which produced decrements in intrinsic motivation relative to control conditions of no reward. This pattern of results was found across five different operational definitions of intrinsic motivation, some of which were behavioral and some of which were self-report, attitudinal measures. Most striking, perhaps, is the fact that this pattern of results was still found 1 month after the experimental manipulation had taken place, on the task enjoyment measure.

The findings of this study lend support to the generalizability of the overjustification effect, which was replicated using a different subject population (high school students as opposed to college sophomores or preschoolers), a different experimental activity, and a different reward (notebook and pens, as opposed to the cash usually offered to adult subjects). The enhancing effect of positive feedback was also replicated, and this effect was found to be independent of the overjustification effect. This finding has important implications for the comparison of previous studies and for the identification of appropriate control groups.

⁵ All reported p levels are two-tailed unless otherwise stated.

Since the receipt of a performance-contingent reward means that a subject has received both a reward *and* positive feedback concerning performance, the appropriate control group for examination of the effects of rewards may be a group that receives the same positive feedback but no reward. The comparison of performance-contingent and task-contingent rewards requires that the task-contingent group receive the same positive feedback, assuming that the feedback does not interact with the offer of a reward. On the other hand, it can be argued that the feedback inherent in a performance-contingent reward is an integral part of the reward itself, in which case it would not be unreasonable to compare performance-contingent rewards with task-contingent rewards or controls without positive feedback added. Choice of an appropriate comparison group is therefore somewhat arbitrary but of course affects the nature of the conclusions drawn.

Both kinds of performance-contingent rewards were found to reduce intrinsic motivation relative to control conditions of no reward. The informational reward (no norms) decreased intrinsic motivation more than the noninformational (norms supplied) reward, as predicted. Although the reward itself contained more information, subjects in both conditions received the same amount of information about their performance. In the noninformational condition, subjects knew how many Ninas they had to find and knew that they were doing well as soon as they found four Ninas. Subjects in the informational condition, on the other hand, had no idea about their performance level until after they finished the puzzles and received their reward. Intrinsic motivation may have been lower in the informational condition because the reward was more controlling (students had to perform as well as possible for the whole session, whereas those in the noninformational condition were less controlled because of the lesser demand of having to find four Ninas), or intrinsic motivation may have been higher in the noninformational condition because of the greater amount of feedback available to subjects during the actual process of doing the puzzles. Positive feedback may be more effective

if it is supplied (or implied by preestablished norms) during the process of task involvement (so that subjects feel that they are doing well) rather than after task completion.

The informational hypothesis of Karniol and Ross (1977) received no support in this study. The means for intrinsic motivation fell in almost the exact opposite ordering of the pattern predicted by their hypothesis. It was originally predicted that their hypothesis would not be supported in this study because of differences in subject populations, but post-experimental conversations with students revealed that the nature of the experimental activity might be an important determinant of the relative salience of controlling versus informational aspects of performance-contingent rewards. Although students enjoyed doing the Nina puzzles, their enjoyment did not seem related to competence concerns. Enjoyment of the task was only slightly related to actual performance level, $r(91) = .22$, $p < .05$. Neither performance nor perceived performance was significantly related to any of the other measures of intrinsic motivation, suggesting that actual or perceived competence is not always an integral part of intrinsic motivation.

Intrinsically motivated activities may be engaged in for their own sake (e.g., drawing with magic markers and reading novels) or for the feelings of competence they produce (e.g., games, puzzles, and sports). This is, of course, an unrealistic dichotomy, because most activities involve both kinds of intrinsic satisfaction, but the extent to which an activity is relatively more competence relevant or fun relevant may determine whether informational or controlling aspects of a performance-contingent reward will be more salient to an individual.

The results of the analyses of recall differences are particularly interesting. The fact that recall was unrelated to the measures of intrinsic motivation and unrelated to performance, yet was affected by reward condition, suggests that this measure tapped an important dimension of task involvement that has not previously been examined in overjustification studies. The finding that contingently rewarded subjects did not remember

as much about the puzzles as did other subjects, but did not find any more Ninas either, suggests that they may have been using a more answer-oriented strategy that did not prove to be effective in terms of performance on these particular puzzles. Exactly what sort of processing strategy is gauged by the recall measure employed in this study is not clear, but the results of the analyses lend support to Condry's (1977) contention that more attention needs to be directed toward an exploration and understanding of the process of task involvement.

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